

Apriori Algorithm - Complete Solution

STEP 1 — SUPPORT THRESHOLD

Total transactions = 5

Minimum support = 60% = 3 transactions.

STEP 2 — TRANSACTIONS

T1: Mango, Onion, Nintendo, Key-chain, Eggs, Yo-yo

T2: Doll, Onion, Nintendo, Key-chain, Eggs, Yo-yo

T3: Mango, Apple, Key-chain, Eggs

T4: Mango, Umbrella, Corn, Key-chain, Yo-yo

T5: Corn, Onion, Key-chain, Ice-cream, Eggs

STEP 3 — FREQUENT 1-ITEMSETS

Counts:

Mango(3), Onion(3), Key-chain(5), Eggs(4), Yo-yo(3)

Frequent (≥ 3):

{ Mango, Onion, Key-chain, Eggs, Yo-yo }

STEP 4 — FREQUENT 2-ITEMSETS

(Mango, Key-chain)=3

(Onion, Key-chain)=3

(Onion, Eggs)=3

(Key-chain, Eggs)=4

(Key-chain, Yo-yo)=3

Frequent:

{ (Mango, Key-chain), (Onion, Key-chain), (Onion, Eggs), (Key-chain, Eggs), (Key-chain, Yo-yo) }

STEP 5 — FREQUENT 3-ITEMSETS

(Onions, Key-chain, Eggs) = 3

Frequent:

{ (Onion, Key-chain, Eggs) }

STEP 6 — ASSOCIATION RULES

Rule 1: (Onion, Key-chain) → Eggs

Support = 60%, Confidence = 100%, Lift = 1.25

Rule 2: (Onion, Eggs) → Key-chain

Support = 60%, Confidence = 100%, Lift = 1.00

Rule 3: Onion → (Key-chain, Eggs)

Support = 60%, Confidence = 100%, Lift = 1.25

FINAL STRONG RULES (Confidence ≥ 80%):

1. (Onion, Key-chain) → Eggs

2. (Onion, Eggs) → Key-chain

3. Onion → (Key-chain, Eggs)

